



SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

Department Of Computer Science and Engineering (AI & ML) Intelligent Placement and Career Development Platform



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ABSTRACT

The aim of this project is to develop a smart and automated intruder detection system using Raspberry Pi that enhances security in restricted areas. The system integrates multiple sensors including PIR, ultrasonic, and sound sensors to detect motion, object presence, and unusual noise. A USB camera captures real-time images of intruders, even in low-light and dark conditions, and sends alerts to the authorized user.

By combining sensor data and image processing, the system reduces false alarms and ensures reliable monitoring. This solution provides an efficient, low-cost, and intelligent alternative to systems.

EXISTING SYSTEM METHODS

- Manual monitoring
- CCTV based surveillance systems
- Single sensor security systems
- Basic alarm and buzzer alerts

DISADVANTAGES OF EXISTING SYSTEM

- Poor performance in dark conditions
- Absence of anomaly detection
- Limited detection with single sensors
- Generates high false alarms

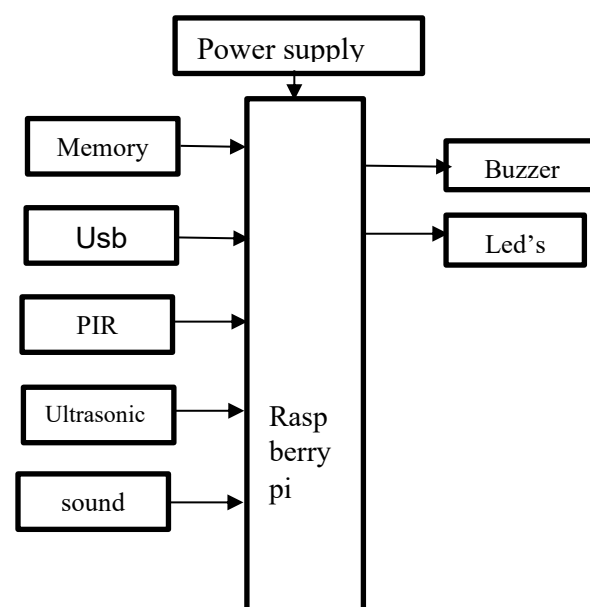
PROPOSED SYSTEM

The proposed system is an IoT Based smart intruder detection system using Raspberry Pi and multiple sensors for accurate monitoring. It performs real-time image capture, face recognition, sound detection and anomaly detection to identify intruders effectively. The system works even in low-light conditions and sends instant alerts while reducing false alarms through the email notifications with the coloured captured images.

ADVANTAGES OF PROPOSED SYSTEM

1. Intruder & anomaly detection in dark and abnormal conditions
2. Instant alerts with coloured images
3. Reduce false alarms

FLOW CHART



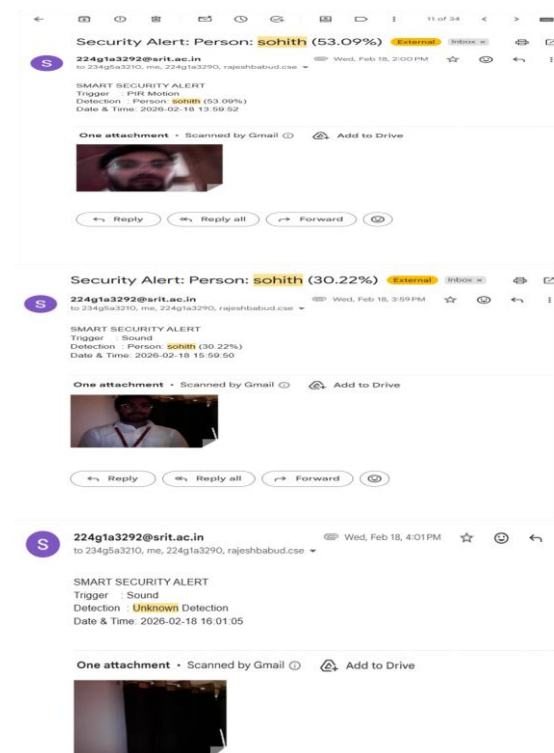
MODULES

1. Motion & sound Detection
2. Face Recognition & Image Capture
3. Alert Generation

MODULES DESCRIPTION

- **Motion & Sound Detection:** Using PIR and sound sensors to detect movement and unusual activity.
- **Face Recognition & Image Capture:** Using USB camera and OpenCV to capture images and identify persons.
- **Alert Generation:** Using buzzer, LED and email to notify the user instantly.

OUTPUTS



CONCLUSION

An IoT Based smart intruder detection system provides an efficient and reliable solution for modern security needs by integrating multiple sensors, image processing, and real-time alert mechanisms. It overcomes the limitations of traditional systems by working effectively in low-light conditions, reducing false alarms, and enabling intelligent detection through face recognition and anomaly analysis. The system ensures quick response and enhanced security through automation, making it suitable for restricted areas.

BATCH NO: B – 07

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